ECDLP lead atlas: first full review (2026-10-24)

Question. Which ECDLP leads have the four repositories (crypto-autoresearcher, crypto, cryptanalysis, ecc-challenges) already examined? What decided each one, and what is left? The aim is a state in which a fresh clone can tell, by computation, that everything has already been examined.

Scope and method. Seven parallel surveys, one per lead family (generic rho, binary IC, the IC solving engine, prime-field IC, transfers, curve structure, and exotic or statistical ideas). They read the summaries in all four repositories: decision and evidence records, findings, problem records, study READMEs and conclusions, leaderboards and benchmark indexes. Each survey drafted lead rows and mapped every ECDLP-area ledger record to a lead. I merged the rows and resolved cross-family duplicates. I checked every row with `tools/lead_atlas.py --check --strict-paths`: every evidence path exists, and every ledger and KR id resolves. I then folded in the cryptanalysis `main` work that the first pass could not see (its cloned default branch is a stale lineage; see Limitations).

Status of each statement. Dispositions are *derived* from committed evidence and cite it. They are not new measurements and change no hypothesis, experiment or goal status. Numbers are *measured* values copied from the cited records. Rankings are this review's *proposal*.

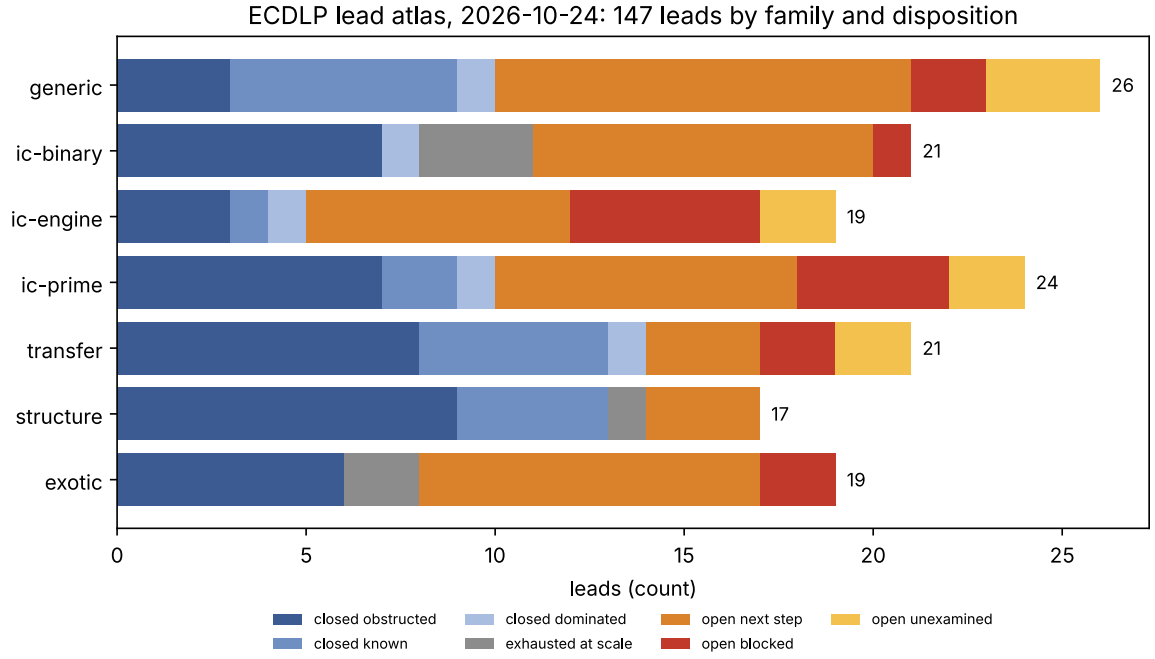


Figure 1: Leads by family and disposition

Result

- **147 leads**: 72 closed, each with a named obstruction (43 obstructed, 18 known, 5 dominated, 6 exhausted at scale), 50 open with a decided next step, 16 blocked and 9 unexamined.
- **Coverage 2049/2049**: every hypothesis and idea in an ECDLP area of the ledger maps to a lead (`knowledge/leads/ecdlp/coverage.tsv`). 264 rows are mapped `NONE` because they are not about the discrete log: rank records, code-based and PQC strays, and arithmetic trivia.
- **Review pinned**: `REVIEW_STATE.json` records the default-branch commit of each repository: autoresearcher `4fa15469d`, crypto `da3a1b5d5`, cryptanalysis `9760a763d`, ecc-challenges `3a71596ba`. `--delta` is empty, so the session-start brief now says that nothing has changed since the review.

family	leads	closed	open, next step	blocked	unexamined
generic (rho, kangaroo, BSGS, pool)	26	10	11	2	3
ic-binary	21	11	9	1	0
ic-engine	19	5	7	5	2
ic-prime	24	10	8	4	2
transfer	21	14	3	2	2
structure	17	14	3	0	0
exotic	19	8	9	2	0

The structure and transfer families are nearly exhausted. The theorems that close them are Tate's isogeny theorem, the Balasubramanian–Koblitz embedding-degree condition, the genus-g index-calculus floor `q^{2-2/g} ≥ q`, and the triviality of homomorphisms from a prime-order group into an ℓ-group (ℓ ≠ the group order). The open frontier is index-calculus engineering and the exponent questions inside the solving engine.

Distance remaining to matched rho (measured, copied)

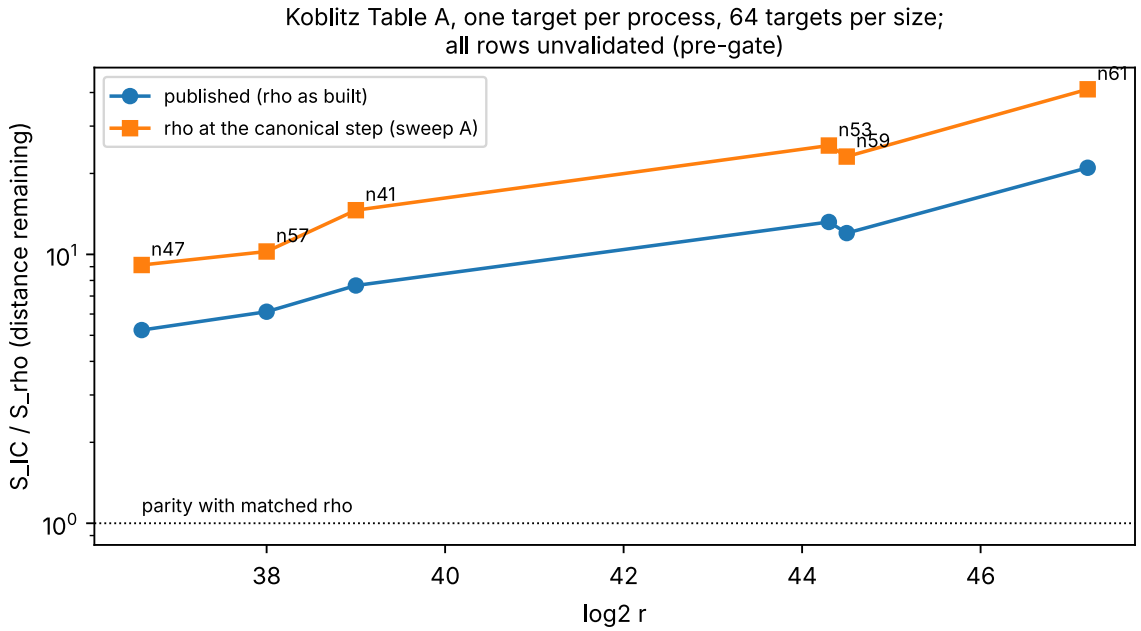


Figure 2: Table A gap, published and at the canonical rho step

board	best row	IC / matched rho	caveat
crypto Table A (Koblitz, one target)	n47	5.23x published, 9.13x at the canonical step	rho priced "as built"; sweep A re-prices at 9.13–41.1x
crypto Table B (prime field)	fp12	5.47x	gap widens with log2 r to 13.5x at fp24
crypto Table C (Koblitz, 32 targets)	n41	3.62x	rho from a different session
cryptanalysis Koblitz ratchet	n83 rank-finder row	3.92x cold vs tuned rho	wall time on one host class, factor-base construction uncharged, §5a band unresolved
generic rho vs floor	Koblitz signed-Frobenius strong rho	1.089x floor	canonicalisation counted, not charged

Every IC row is still "unvalidated (pre-gate)". No row is below its reference. The online "IC beats rho" readings (n61 13.9x, N83 18.5x and similar) compared IC against rho with an empty distinguished-point table. They are not results: at equal advice, Bernstein–Lange rho is 53–317x faster online.

Next steps, ranked across families (proposal)

1. **Port the cryptanalysis n=83 engineering ratchet to crypto Table A, then re-run Table A** with tuned K and production free relations (`LEAD-ICB-koblitz-setup-phase-cost`). Set-up is 98.5–99.7% of Table A's S. Off-board figures (K retune 0.46x at n41, free relations 0.960x) and the ten stacked n=83 producers indicate that this is the cheapest verified IC-baseline improvement. It is constant-level only.
2. **Validation gate for the IC boards.** No row can land until it carries a replay certificate, a validator report and the §5a band (crypto CLAUDE.md, gate of 2026-10-19). This blocks every other improvement from counting.
3. **Descent closure degree at (m, l) = (3, 7) and (4, 7)** (`LEAD-ICE-descent-closure-degree-at-scale`, blocked on a host with at least 250 GB). This is the only exponent-level question left for prime-degree binary curves: degree 4 puts the ECC2K-130 m=4 row about +7.6 bits over rho, degree 5 about +10.3 bits, and degree 6 or more closes it. The new `LEAD-ICE-hybrid-guess-determine-l7` probes the same question without that host.

4. **Class quotient in the ecc-challenges v2 worker** (`LEAD-RHO-pool-class-quotient`). Negation (√2) and, on Koblitz campaigns, signed Frobenius (√(2m)) are missing from all 23 non-ECC2K-130 campaigns. The win is certain engineering.
5. **Price the native-XOR SAT arm** (`LEAD-ICE-sat-xor-pdp-oracle`). Its S/rho of 0.854 at n=17 Koblitz holds only because solver conflicts are unpriced. One calibration decides the lead either way.
6. **Optimal Coppersmith shift set** (`LEAD-ICP-coppersmith-smallroot-semaev`). The recorded closure "by 0.048" used one Jochemsz–May strategy, with 0.159 of slack unused. Ryan's (EUROCRYPT 2025) optimal shifts at (m, w) = (4, 2) are a calculator run with a decisive outcome.
7. **Joux–Vitse sieve on the GF(p^6) weak class, with full accounting** (`LEAD-TRF-jv-sieve-fp6-weak-class`). This is the only transfer ever measured below strong rho (0.011x including the walk), but its comparisons were withheld because the tolerances and hash costs were not frozen. It reproduces a 2012 result and applies to the weak class only.
8. **Exotic lanes with a decisive cheap test**: the bucket variance W of the four-list O2 kernel (`LEAD-EXO-four-list-ktree-join`), and claims K1 and K2 of the Miller cocycle (`LEAD-EXO-miller-pairing-cocycle`).

New leads added by this review (`open_unexamined`): `LEAD-ICE-linearization-excess-fb-prior`, `LEAD-ICE-hybrid-guess-determine-l7` and `LEAD-TRF-teske-trapdoor-fp6-weak-class`. The surveys proposed nine more candidates; `--match` found each one already present as a lead or a KR row. That is the gate doing its job.

Integrity findings for the Coordinator (observed; no status changed here)

- **Ledger backlog.** 971 of the 1,031 ECDLP-area hypotheses (94%) have no terminal status: 373 approved, 308 specified, 240 proposed, 47 analyzed, 3 other. The surveys counted more than 100 approved hypotheses that are arithmetic trivia or Stage-0 identity checks ("Eight times six is 48"). They inflate counts without touching a lead.
- **Stale or contradicted documents an agent would re-propose from:**
  - `crypto/reports/Index calculus leads next steps.md` calls ten items unexecuted that `research/ic_next10_20261014/` ran.
  - `analysis/o2-sum-compatible-filters/README.md` says the four-tree route is closed; KN-FIND-ffe1df says it is not.
  - Two monodromy findings conflict: KN-FIND-a1f3c2 against KN-FIND-bdcd60.
  - Two elliptic-net findings conflict: KN-FIND-002 against KN-FIND-b7e091.
- **Status disagrees with evidence:**
  - H-IT-001 was closed on Tate's theorem (EV-IT-511f3d, strong), then reset to `weakened` with no new evidence.
  - H-ENDO-001 is `approved` against strong contradicting evidence.
  - H-QSP-cd0c90 expired with no result but is still `approved`.
- **Results without evidence records:** EXP-GRUMPY-cd7b82 stage 1, RUN-AUXIN-fd1edc, five KIC runs and the CZLIFT round (20 validated trials).
- **Instruments that could not fail:**
  - The XEDN `max|coeff| = 1` statistic.
  - ECTD meters: identical on all 32 curves.
  - The volcano census's ±10% IC margin. It could not see the measured 5.8% ℓ=2 floor deficit.
- **Overbroad closure:** KR-RHO-755e35 and EV-IT-511f3d say that isogeny walks never find weak curves. That is true over prime fields and false over composite-degree binary fields, where it is exactly how the GHS walks and Teske's trapdoor work.
- **Mislabelled regression:** the leaderboard's "96 root-finder replay mismatches" are inherited-F4 `solver_ops` rows (BMI2 rewrite, F4 pricing changes), per validation sweep D. They are not root-finder rows.
- **Repository hygiene:** the cryptanalysis GitHub default branch is a stale lineage (`claude/cryptanalysis-repo-setup-p59kek`, about 530 commits not on `main`). Sessions cloning the default read old results. Four atlas evidence paths exist only there, so they are pinned with `ref: c273e621`.

How the harness uses this

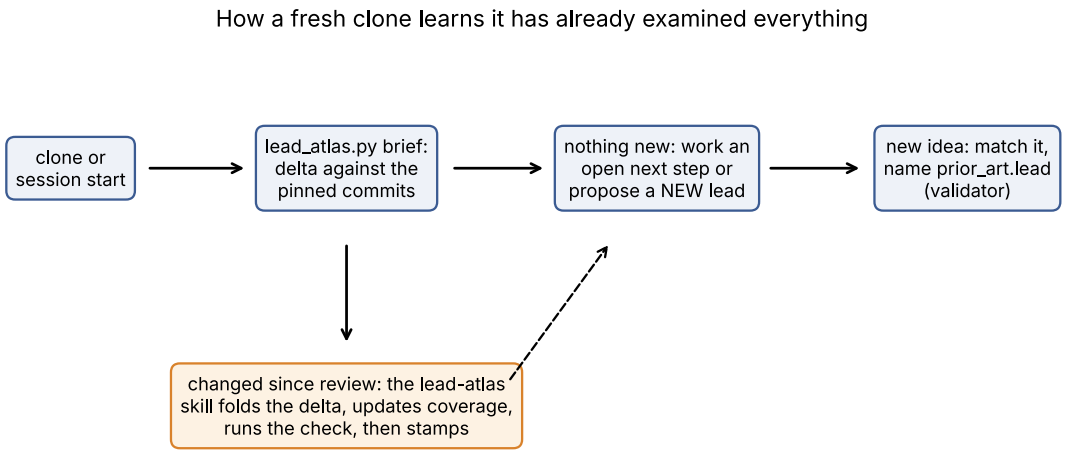


Figure 3: Harness loop

- **Session start.** All three research repositories print `tools/lead_atlas.py --brief`. When `--delta` is empty, the brief says that everything was already examined and points at the open next steps.
- **Ideation.** `/propose-ideas` pastes `--render` and runs `--match` on each returned idea. From IDEA-20261025 on, the validator requires `prior_art.lead`: the lead the idea advances, or `new` with a reason.
- **Review.** The `lead-atlas` skill folds a non-empty delta into leads, then runs `--check` and `--stamp`. CI and `make check-ledger` run `--check`.
- **Saturation.** No unexamined lead, no unmapped record, and nothing new since review. Today two of the three gates hold: every record is mapped and nothing has changed since the review. Nine leads remain unexamined. Saturation is reported, never used to pause a goal.

Limitations

- **Survey depth.** The surveys read summaries, not every raw run. A disposition is only as good as the decision or evidence record it cites. The coverage mapping is a scripted classification with hand overrides, and the boundary between `OTHER:ic-prime` and exotic among RQ-ECDLP-002 Stage-0 records is noisy.
- **Cryptanalysis lineage.** The first pass read cryptanalysis at its stale default branch. The `main`-only ECDLP material was folded in afterwards: the n=83 IC ratchet, the genus-2 cover and speed studies, and the n=83 surfaces run. Tooling-only changes were not.
- **Unvalidated sources.** Every gap figure is copied from unvalidated (pre-gate) rows.

Sources: the lead rows (`knowledge/leads/ecdlp/<family>/`), each citing its own records; the snapshot of all rows (`atlas-snapshot.md`); figures (`figures.py`, drawn only from committed rows and the cited leaderboard and sweep values).